



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

DIGITAL FORENSICS

System Forensics, Investigation, and Response

Lesson 5

Understanding Techniques for Hiding and Scrambling Information

Learning Objective

- Explain techniques for hiding and scrambling information as well as how data is recovered.

Key Concepts

- Steganography
- Encryption

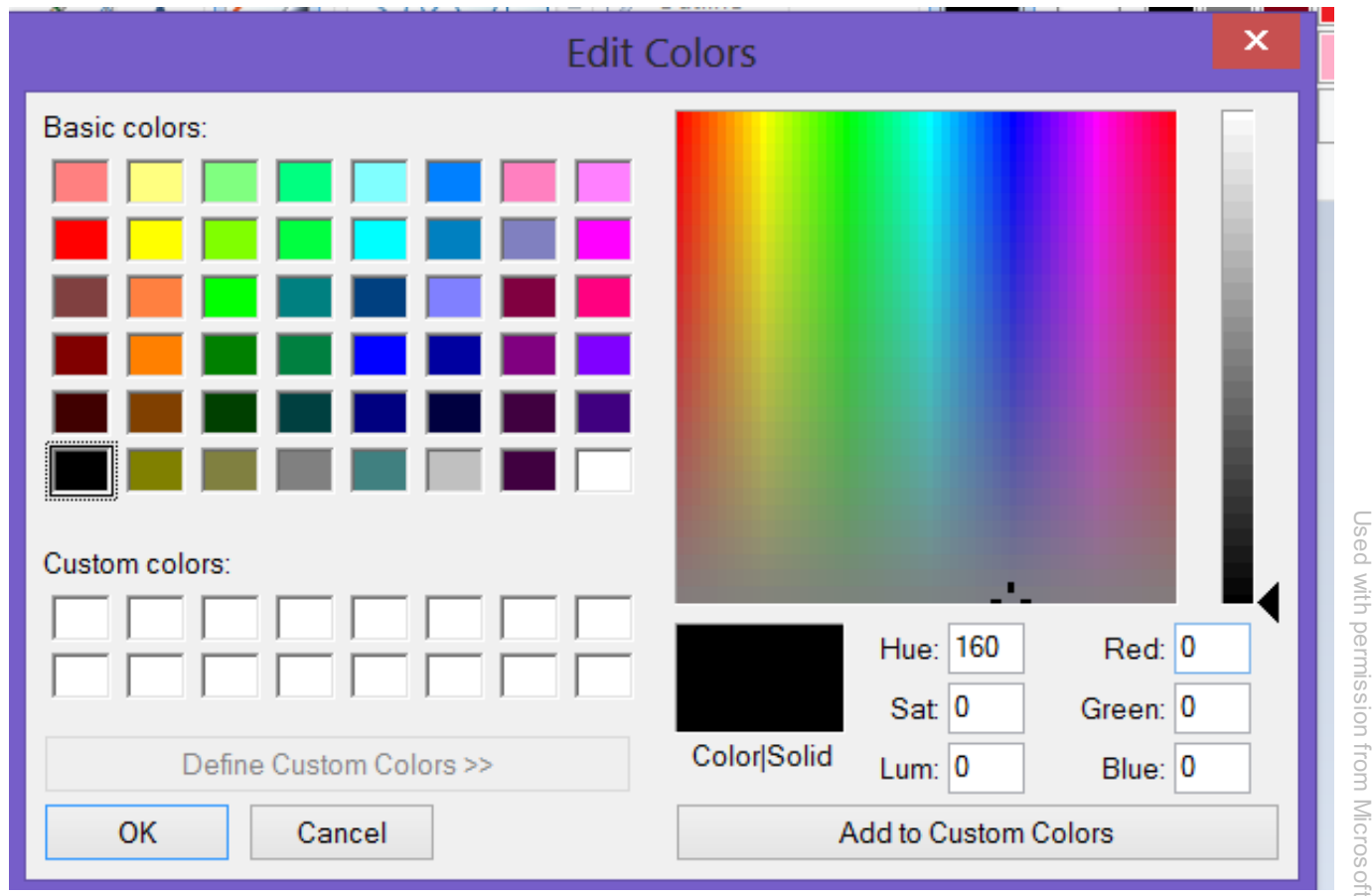
Steganography

- The art and science of writing hidden messages
- Goal is to hide information so that even if it is intercepted, it is not clear that information is hidden there
- Most common method today is to hide messages in pictures using the least significant bit (LSB) method

LSB Method

- Consider 11111111
- Change last digit to 0
- 11111110 = 254 in decimal
- The last bit or least significant bit is used to store data
- Colored pixels in a computer stored in bits

LSB Method



LSB Method

Color|Solid

Hue: 0 Red: 252

Sat: 231 Green: 101

Lum: 166 Blue: 100|

Color|Solid

Hue: 0 Red: 252

Sat: 231 Green: 100

Lum: 166 Blue: 100|

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Basic Steganography Terms

Payload

Carrier

Channel

Historical Steganography

- Ancient Chinese wrapped notes in wax and swallowed them for transport
- In ancient Greece, message written on slave's shaved head, then hair allowed to grow back
- During World War II, French Resistance sent messages written on the backs of couriers using invisible ink

Other Forms of Steganography

Steganophony

- Hiding messages in sound files

Video steganography

- Hiding information in video files

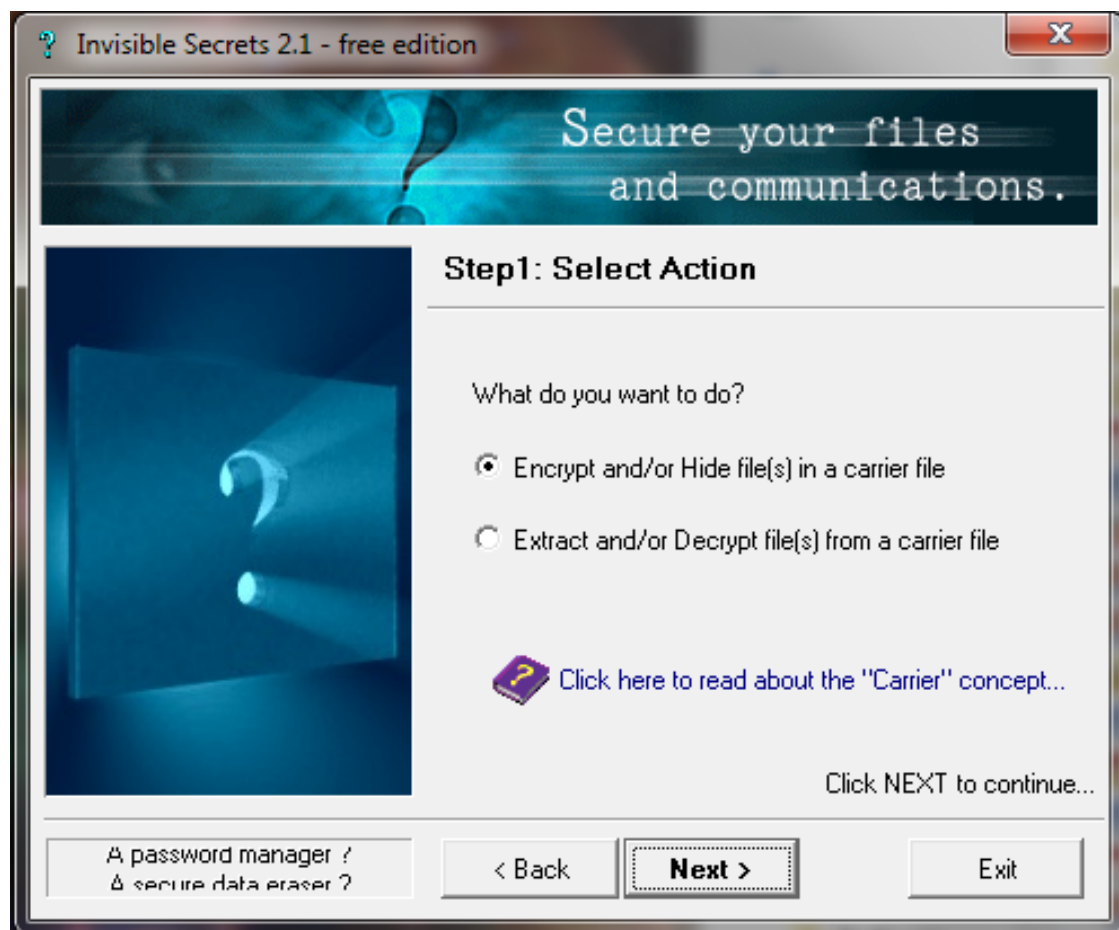
Advanced steganography

- Bit-plane complexity segmentation steganography (BPCS)

Steganalysis

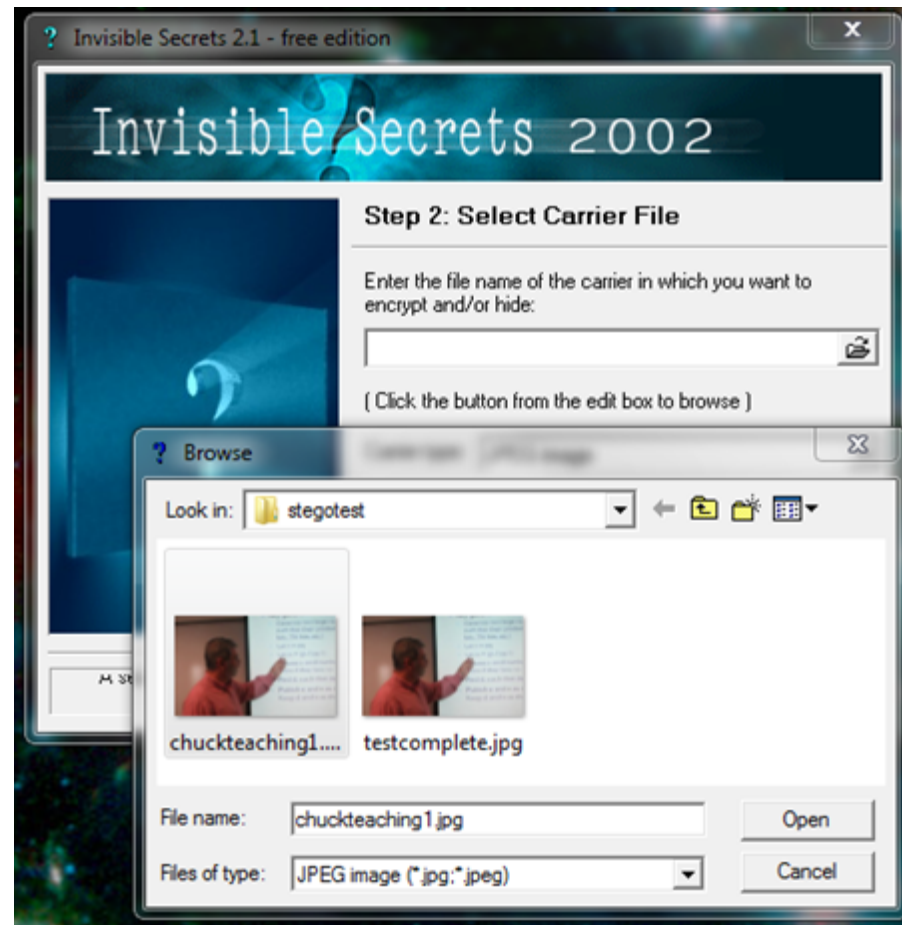
- The process of analyzing a file or files for hidden content
- Can show a likelihood that a given file has additional information hidden in it
- Common method for detecting LSB steganography is to examine close-color pairs

Invisible Secrets



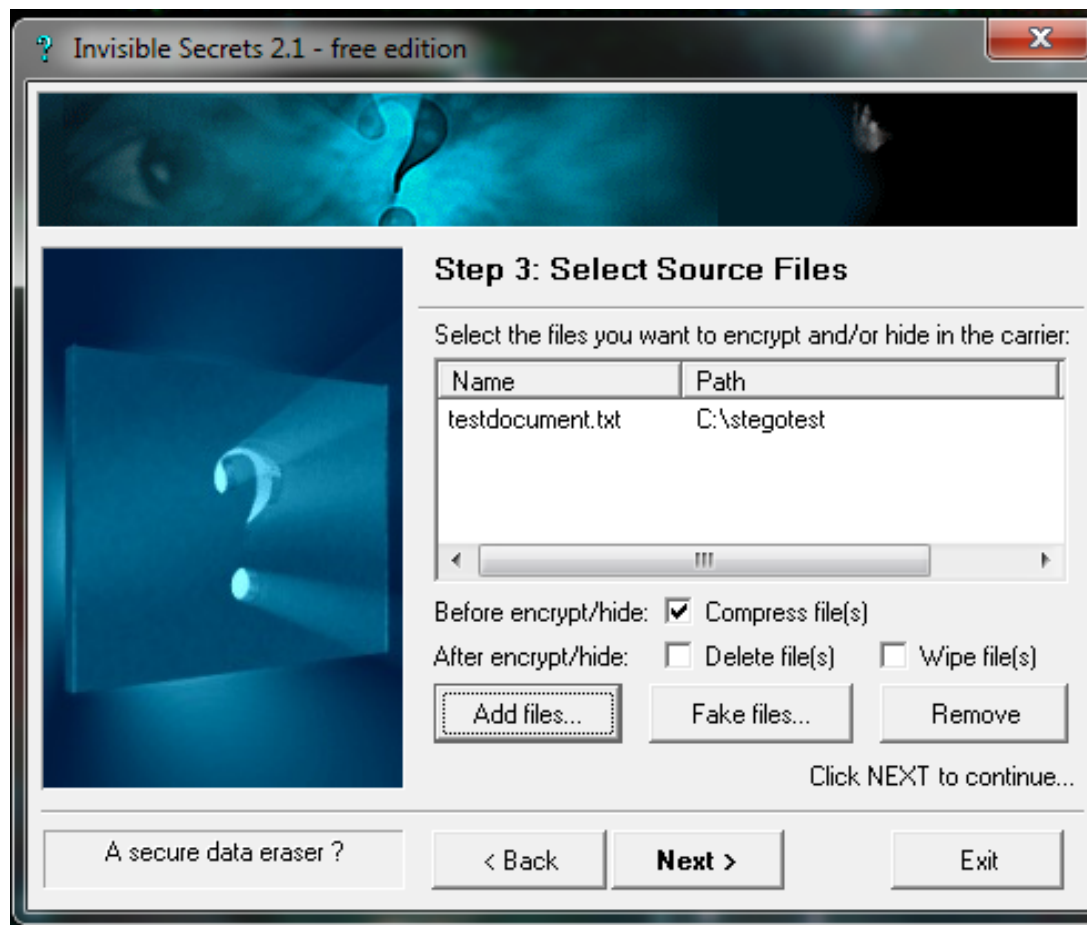
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Invisible Secrets



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Invisible Secrets



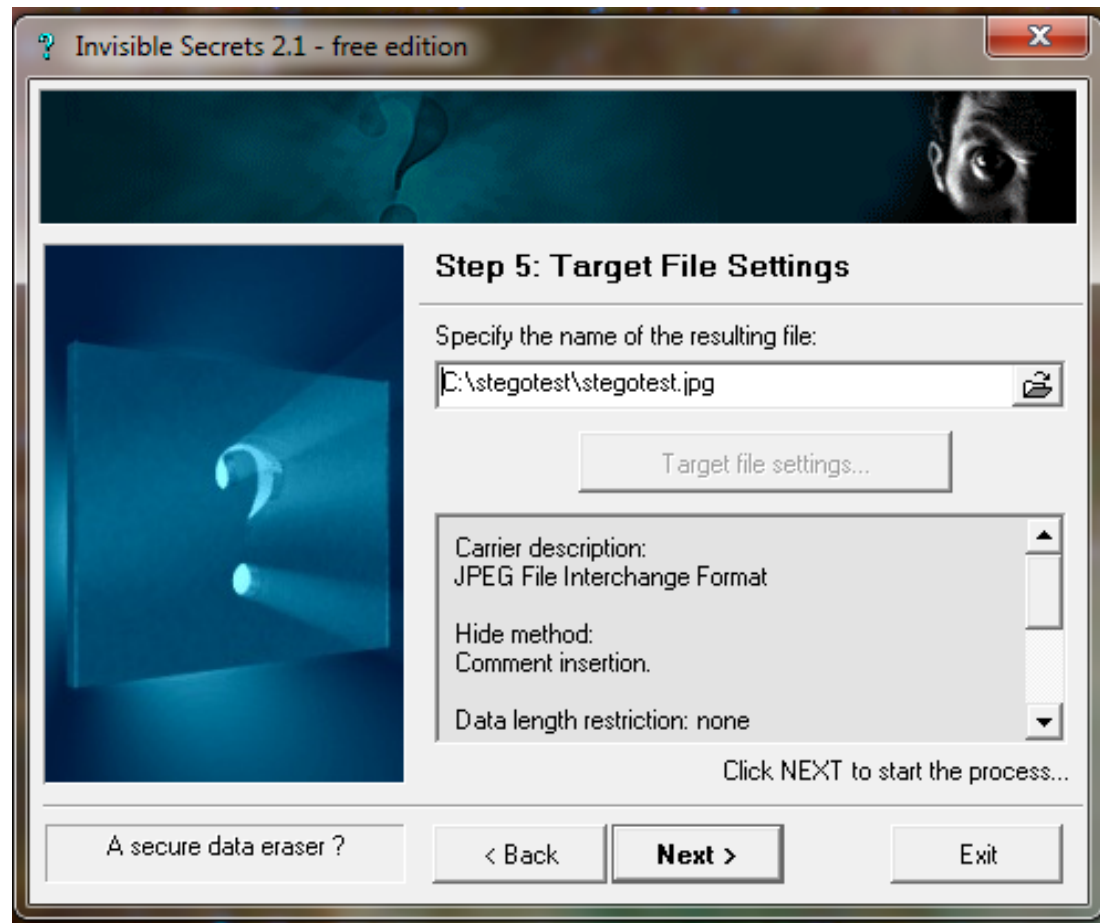
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Invisible Secrets

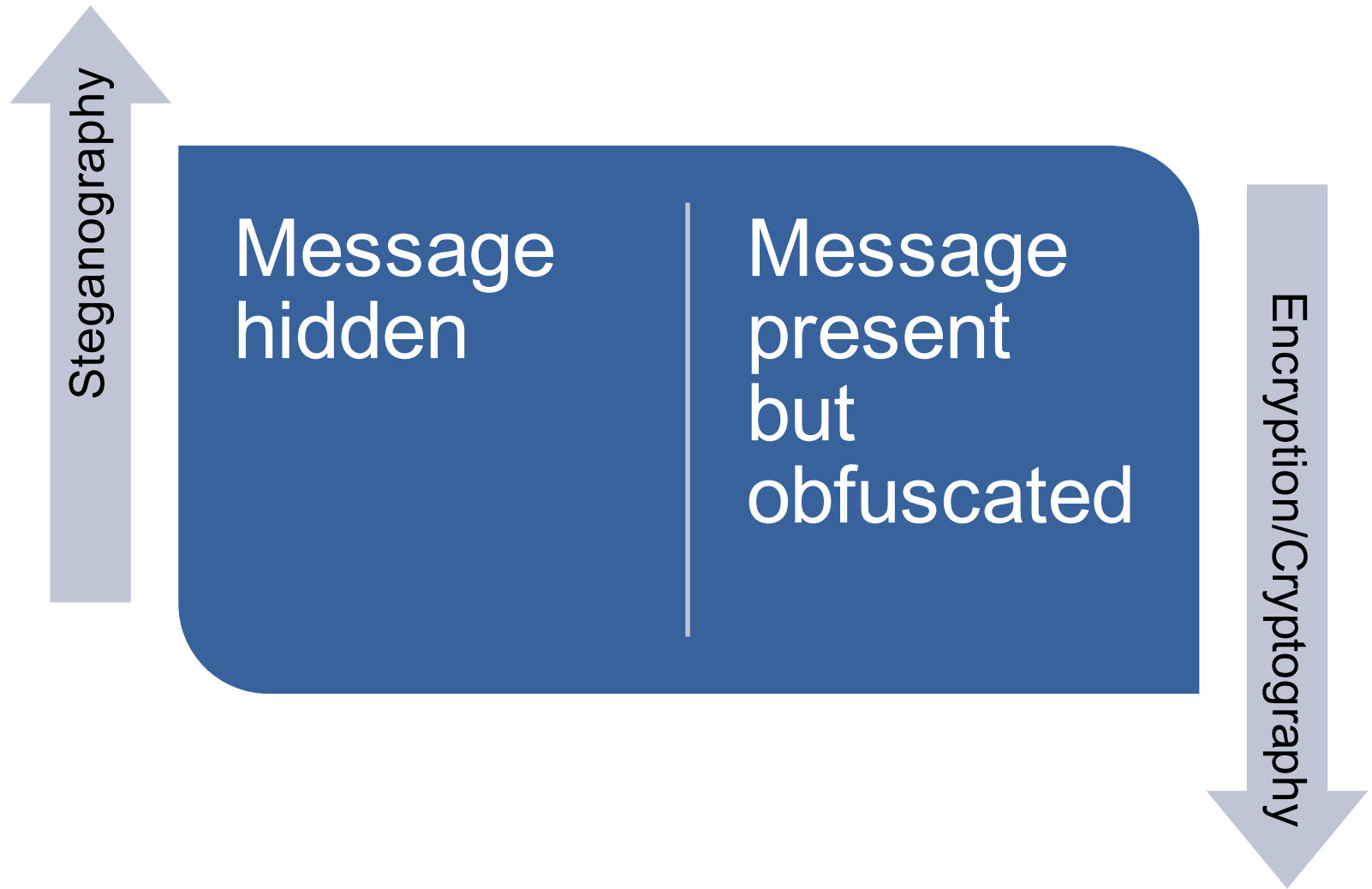


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Encryption

- Encryption obfuscates message so it cannot be read
- Cryptography is the study of writing secret messages
- The word cryptography derived from word *kryptós*, which means hidden, and the verb *gráfo*, which means write

Encryption vs. Steganography



The Caesar Cipher

- Referred to as the substitution cipher
 - A simple method of encryption and very easy to crack
-
1. Choose some number by which to shift each letter of a text
 2. Substitute the new alphabetic letter for the letter being encrypted

Caesar Cipher Example

- Text is:
 - A CAT
- You choose to shift by two letters, then A replaces C, E replaces C, C replaces A, and V replaces T; encrypted message is:
 - C ECV
- If shift by three letters, message is:
 - D FDW

The Caesar Cipher

- Introduced two key terms:
 - Text you want to encrypt is referred to as *plaintext*
 - After it has been subjected to the algorithm and key, resultant text is called *ciphertext*

The Atbash Cipher

- Reverses the alphabet—substituting the first letter of the alphabet for the last letter, the second letter for the second-to-last letter, and so on
- Is primitive and easy to break

The ROT13 Cipher

- A permutation of the Caesar cipher
- All characters are rotated 13 characters through the alphabet
- A CAT becomes N PNG

The Scytale Cipher

- Encrypts messages by wrapping a leather strip around a cylinder or baton, and writing across the leather
- Turning cylinder produced different ciphertexts
- Message decrypted by reading the message once placed over the same leather “key” wrapped around the same size cylinder

Playfair

- Works by encrypting pairs of letters, also called digraphs, at a time
- Uses a 5×5 table that contains a keyword or key phrase
- To use the Playfair cipher, one need only memorize that keyword and four rules
- Example: “Attack at dawn” becomes "At ta
ck at da wn"

Playfair (Cont.)

<u>F</u>	<u>A</u>	<u>L</u>	<u>C</u>	<u>O</u>
<u>N</u>	<u>B</u>	<u>D</u>	<u>E</u>	<u>G</u>
<u>H</u>	<u>I/J</u>	<u>K</u>	<u>M</u>	<u>P</u>
<u>Q</u>	<u>R</u>	<u>S</u>	<u>T</u>	<u>U</u>
<u>V</u>	<u>W</u>	<u>X</u>	<u>Y</u>	<u>Z</u>

Playfair (Cont.)

<u>A</u>	<u>L</u>	<u>C</u>
<u>B</u>	<u>D</u>	<u>E</u>
<u>I/J</u>	<u>K</u>	<u>M</u>
<u>R</u>	<u>S</u>	<u>T</u>

Playfair (Cont.)

<u>A</u>	<u>L</u>	<u>C</u>
<u>B</u>	<u>D</u>	<u>E</u>
<u>I/J</u>	<u>K</u>	<u>M</u>

Multialphabet Substitution

- Uses multiple numbers by which letters in plaintext are shifted
- Multiple substitution alphabets are created
- Represents a slight improvement on the Caesar cipher but is still easily cracked

Multialphabet Substitution Example

- If you select three substitution alphabets (+2, -2, +3)
- A CAT becomes C ADV

The Vigenère Cipher

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

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The Enigma Machine

- World War II, electromechanical rotor-based cipher system
- Is a multialphabet substitution cipher using machinery to accomplish the encryption
- When operator pressed a key, encrypted ciphertext for plaintext was altered each time

Modern Cryptography

- Two main types: Symmetric and asymmetric
- Symmetric cryptography uses the same key to encrypt and decrypt plaintext
- Asymmetric cryptography uses different keys to encrypt and decrypt plaintext

Modern Cryptography (Cont.)

- Used every day by millions of consumers on the World Wide Web to buy products and services securely
- “https” at beginning of Web address or a padlock symbol indicates a secure protocol such as Transport Layer Security (TLS) is at work

Modern Cryptography (Cont.)

- Cryptography also used in:
 - Antivirus software
 - Wireless security (WPA and WPA2 encryption)
- Hard disk encryption using Microsoft Encrypting File System (EFS) is a form of cryptography
- Did you know your mobile phone transmissions are encrypted, as are your ATM and credit cards?

Symmetric Cryptography

- Uses the same key to encrypt and decrypt plaintext
- Can use two different encryption keys, one from sender to receiver and one from receiver to sender
 - Same key is still used for encryption and decryption
 - Having different keys in both directions provides additional security if keys are learned or disclosed

Substitution and Transposition

- Swapping of blocks of ciphertext
- All modern block-cipher algorithms use substitution and transposition
- Combination of substitution and transposition increases security of resultant ciphertext by making cryptanalysis more complex

Block Ciphers and Stream Ciphers

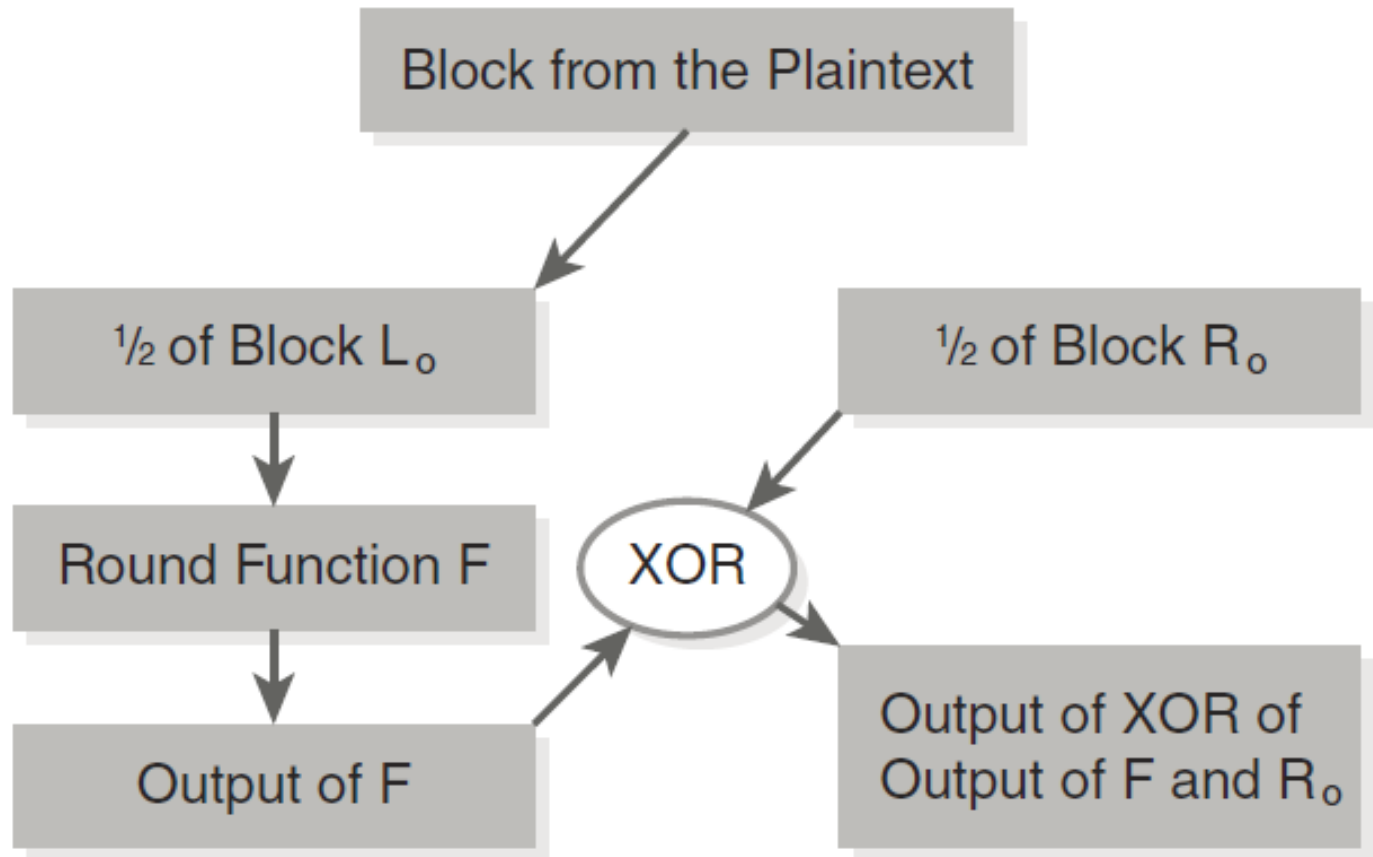
Block cipher

- Encrypts data in groups of bits, also known as blocks

Stream cipher

- Encrypt data as a stream, one bit at a time

The Feistel Function



Data Encryption Standard (DES)

1. Data is divided into 64-bit blocks.
2. Data is manipulated by 16 separate steps of encryption involving substitutions, bit-shifting, and logical operations using a 56-bit key.
3. Data is then further scrambled using a swapping algorithm.
4. Data is transposed one last time.

What Happened to 2DES?

- 2DES basically does DES two times
- Was not much more secure than DES
- Took more time and computer resources to implement
- Not widely used

Triple DES (3DES)

- Was created as an interim solution to DES
- Does DES three times, with three different keys

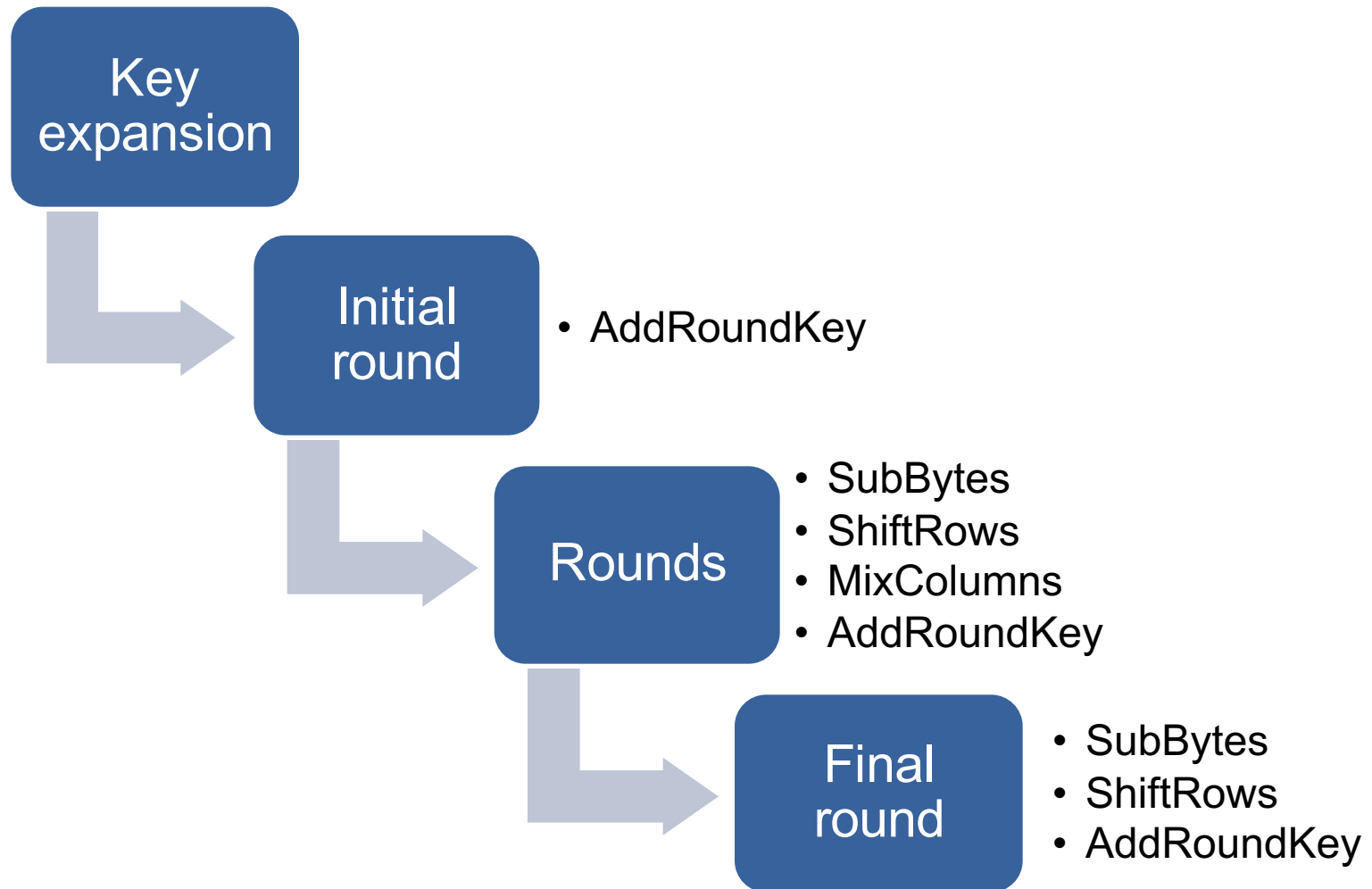
Why 4DES Was Never Implemented

- Early simulations indicated it was too scrambled
- Blocks of original plaintext appeared in the final ciphertext
- One of the driving factors behind searching for a new algorithm not in the DES line

Advanced Encryption Standard (AES)

- Also known as the Rijndael block cipher
- Can have three different key sizes:
 - 128, 192, or 256 bits
- Referred to as AES 128, AES 192, and AES 256

AES Steps



Cryptographic Hashes

- Hashing is a type of cryptographic algorithm with some specific characteristics
 - It is one-way, not reversible
 - It is a fixed-length output no matter what input is given
 - The algorithm must be collision resistant

Asymmetric Cryptography

- Cryptography wherein two keys are used:
 - One to encrypt the message
 - Another to decrypt it

RSA

- Described in 1977 by Ron Rivest, Adi Shamir, and Leonard Adleman at MIT
- Perhaps the most widely used public key cryptography algorithm today
- Is based on relationships of prime numbers
- Security of RSA derives from fact that it is difficult to factor a large integer composed of two or more large prime factors

Diffie-Hellmen

- A cryptographic protocol that allows two parties to establish a shared key over an insecure channel
- Often used to allow parties to exchange a symmetric key through some insecure medium, such as the Internet
- Enabled all secure communications between parties that did not have a pre-established relationship, such as e-commerce

Diffie-Hellmen (Cont.)

- Groundbreaking research provided the foundation for secure transactions across the Internet
- E-commerce sites like Amazon.com and Staples.com can provide secure electronic communications, thanks in great part to Diffie and Hellman



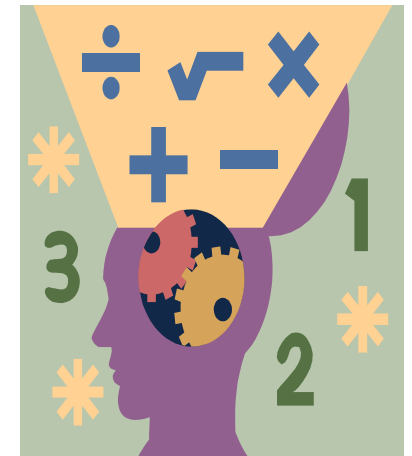
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Breaking Encryption

- Cryptanalysis is using techniques other than brute force to attempt to uncover a key
 - Also referred to as academic or knowledge-based code breaking
- Cryptographic techniques may be used to test the efficacy of a cryptographic algorithm
 - Such as to test hash algorithms for collisions

Frequency Analysis

- In natural languages, certain letters of alphabet appear more frequently than others
- By examining frequencies, can derive some information about the key used
- Method effective against classic ciphers, but not modern methods of cryptography



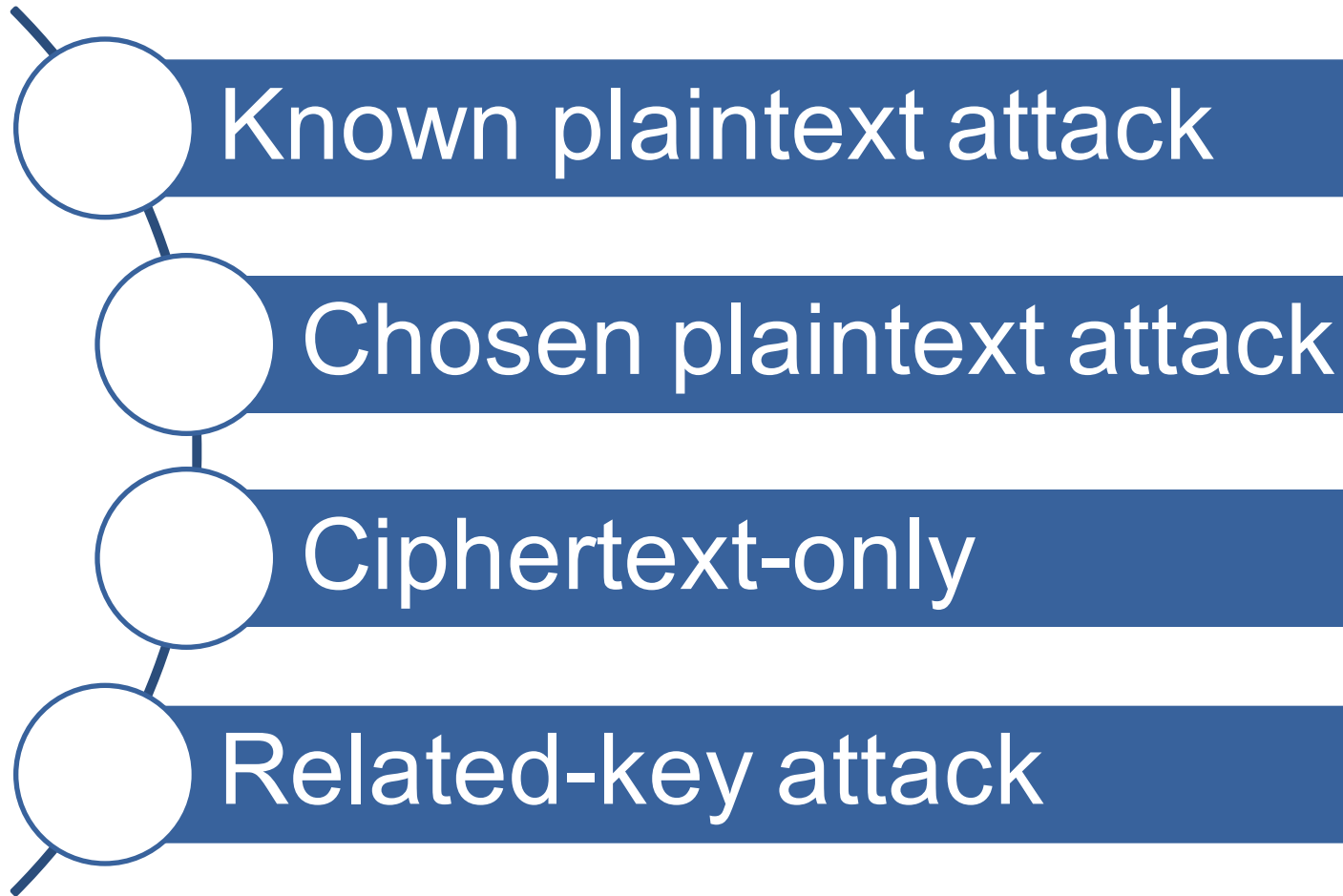
Kasiski

- A method of attacking polyalphabetic substitution ciphers, such as Vigenère
- Can be used to deduce the length of the keyword used in a polyalphabetic substitution cipher

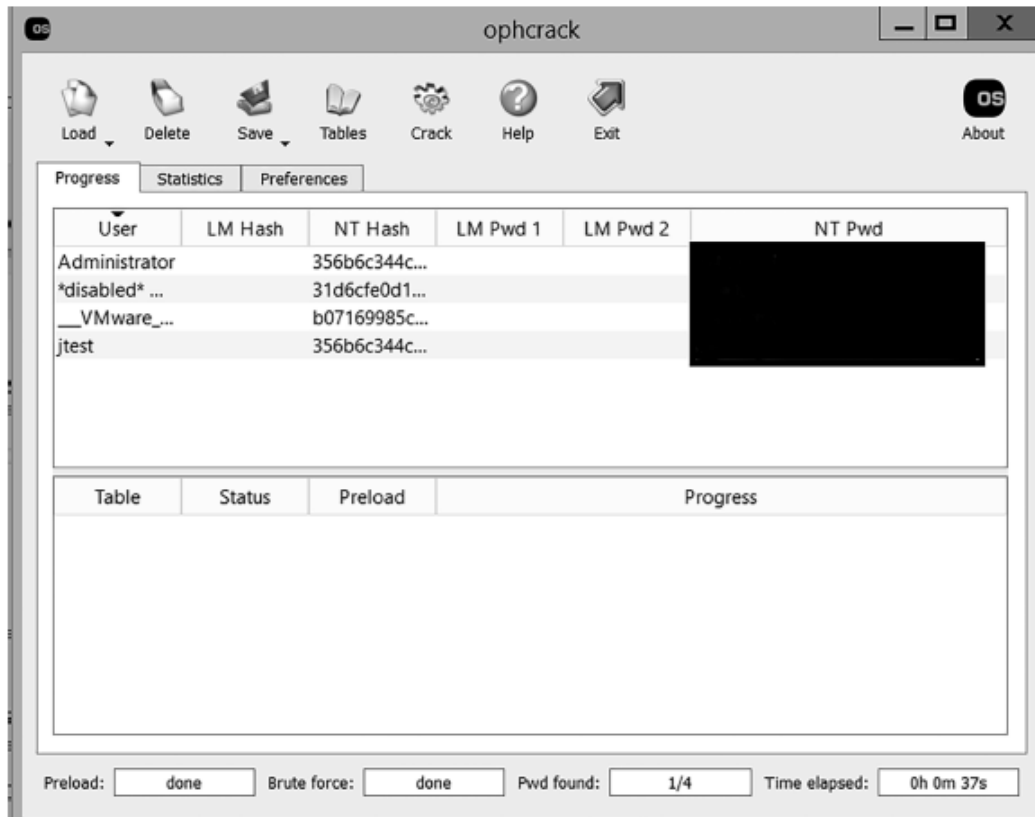
Kasiski (Cont.)

- When length of keyword is discovered, ciphertext is lined up in n columns, where n is keyword length
- Each column
 - Treated as a monoalphabetic substitution cipher
 - Can be cracked with frequency analysis
- Involves looking for repeated strings in the ciphertext

Modern Methods



Rainbow Tables/Ophcrack

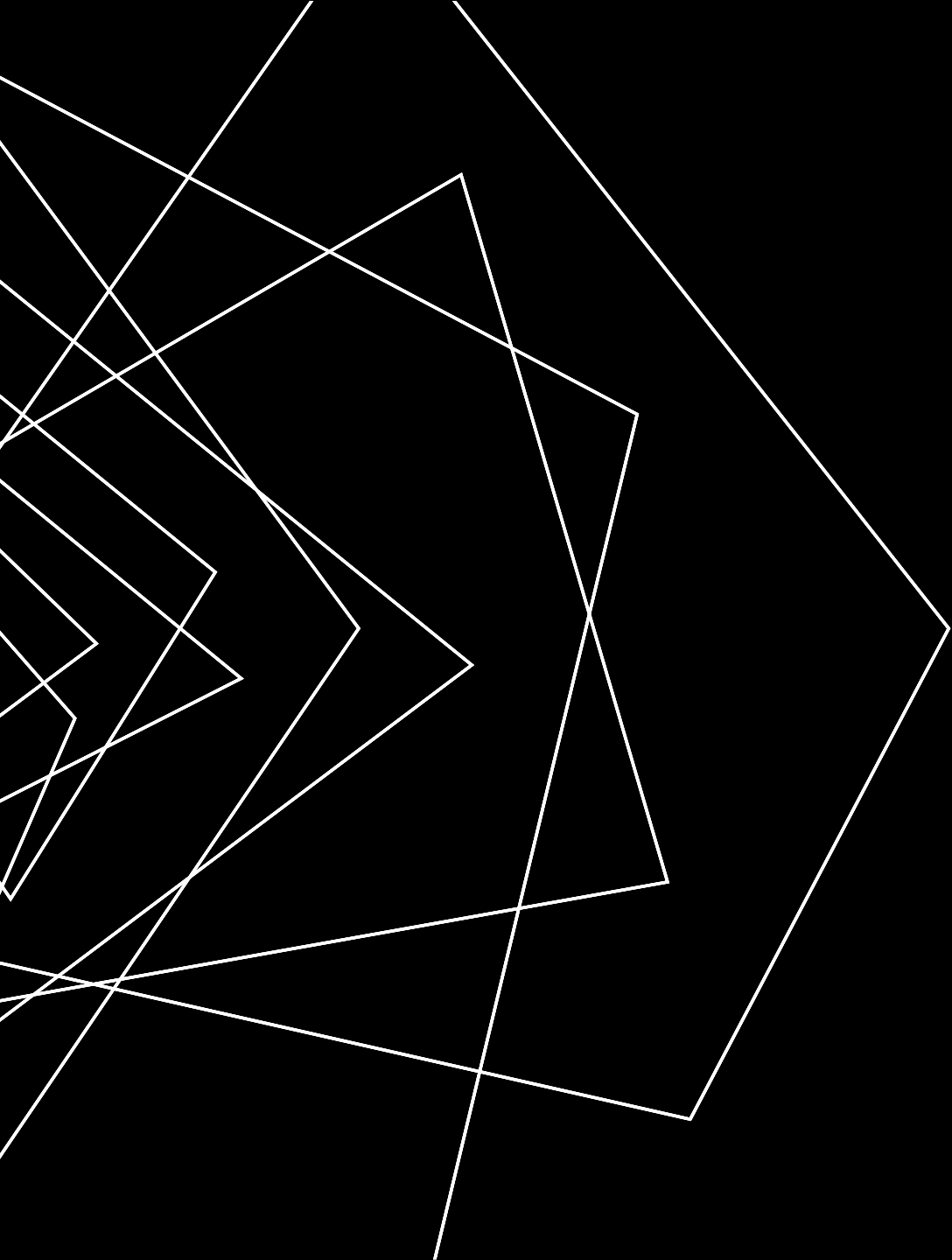


John the Ripper

- A password cracker popular with network administrators and hackers
- Enables user to select text files of word lists to attempt cracking a password
- Command-line based, no Windows interface

Summary

- Steganography
- Encryption



THANK YOU

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